

M002 Extension 5 — Redstone AND Solver

Topic: Redstone AND Rules + Single Checks · **Course:** Maze Madness · **Level:** Extension (Intermediate) · **Time:** about 45 minutes

Same **Week 3 world**. Find the **hardest maze** your teacher shows a picture of, packed with redstone. The agent reads **redstone** to move; tiles with **two** redstones need an **AND** (both true).

The rules you need

- RS **left** → turn right · RS **right** → turn left (single)
- RS **below** → move up 1 (single)
- **#1** RS left **and** below → up 1, turn right (AND)
- **#2** RS right **and** below → up 1, turn left (AND)
- **#4** RS left **and** right → up 5 (AND)

Chat: **l** turn left · **r** turn right · **run** start · **r1** bring the agent back.

1 · Read the Chat Commands

A chat word binds to an action. Type the word, the agent runs the code.

```
def turn_left():
    agent.turn(LEFT_TURN)
player.on_chat("l", turn_left)
```

You type **l**. What does the agent do?

```
def go_back():
    agent.teleport_to_player()
player.on_chat("r1", go_back)
```

You type **r1** . Where does the agent go? When is that handy?

2 · Trace the Solver 🔍

This loop runs when you type `run`.

```
while True:
    if agent.detect(REDSTONE, LEFT) and agent.detect(REDSTONE, RIGHT):
        agent.move(UP, 5)
    elif agent.detect(REDSTONE, LEFT) and agent.detect(REDSTONE, DOWN):
        agent.move(UP, 1)
        agent.turn(RIGHT_TURN)
    agent.move(FORWARD, 1)
```

Rule #4 is `RS LEFT and RS RIGHT`. Which line runs there, and how far does the agent climb?

Rule #1 is `RS LEFT and RS DOWN`. Which line runs, and what two things does it do?

A tile has redstone on the left only — nothing below, nothing on the right. Does either `if` run? Why not?

3 · Fill the Blanks

Rule #2: move up 1 and turn left **only when both are true** — redstone right and redstone below.

```
if agent.detect(REDSTONE, _____) _____ agent.detect(REDSTONE, DOWN):  
    agent.move(UP, 1)  
    agent.turn(_____)
```

Word bank: RIGHT · and · LEFT_TURN

4 · Spot the Bug 🐛

Bug A — Rule #1 should fire **only** when redstone is left **and** below. Right now it checks redstone left only, so it fires on a plain left marker with nothing below it.

```
if agent.detect(REDSTONE, LEFT):  
    agent.move(UP, 1)  
    agent.turn(RIGHT_TURN)  
agent.move(FORWARD, 1)
```

Hint: add the second check with `and` .

Write the fix:

One line: why was it wrong?

5 · Show Your Work 📷 🎥

Open the Week 3 world, find the hardest maze from the picture, and run your solver until the agent reaches the **diamond block** goal.

Record **one video** (a phone is fine). Show two things:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

Write your lines here, then say them in your video.

Submit ✅

Send this worksheet + your video to teacher on KakaoTalk.